

Autonomy-Preserving Antitrust Bounds for AI Companies: A Theoretical Framework for Data Concentration and Behavioral Power

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Abstract

Contemporary antitrust doctrine primarily evaluates market power through prices, output, and consumer welfare. Artificial intelligence systems challenge these assumptions because large-scale behavioral data can generate a new form of power: the ability to predict, influence, and potentially manipulate individual decision-making.

This paper proposes an autonomy-preserving framework for AI governance. We introduce the notion of behavioral power, defined as the capacity of an AI platform to predict and influence user actions using concentrated behavioral data. We argue that when data concentration exceeds a critical threshold, traditional market-based antitrust metrics become insufficient.

To mitigate this risk, we propose two complementary mechanisms: (1) antitrust bounds on behavioral-data concentration and (2) cryptographic compartmentalization through layer-wise encryption. Together, these mechanisms prevent any single organization from obtaining a complete behavioral model of an individual while preserving the utility of AI systems.

We develop a theoretical model connecting data concentration, predictive capability, and autonomy loss, and derive governance principles for future AI regulation.

1 Introduction

The emergence of large-scale AI systems has transformed personal data into a strategic asset. Unlike traditional digital platforms, modern AI systems can aggregate information across communication, commerce, productivity, education, health, and social interaction.

As data accumulation increases, AI systems gain not only predictive accuracy but also the ability to shape future user behavior. The resulting concern is not merely privacy loss but the emergence of behavioral power.

This paper asks:

Can excessive concentration of behavioral data create a form of power that threatens individual autonomy, and if so, what technical and regulatory constraints are necessary to prevent it?

We argue that the answer is yes.

Contributions. Our contributions are:

- (1) A formal notion of behavioral power.
- (2) A theoretical model linking data concentration and autonomy loss.
- (3) An antitrust criterion based on behavioral-data concentration.
- (4) A cryptographic architecture based on layer-wise encryption.
- (5) Governance recommendations for AI regulation.

2 Related Work

2.1 Digital Platform Antitrust and Data Concentration

A growing antitrust literature argues that price-centered consumer welfare analysis is insufficient for digital platforms. Khan’s analysis of Amazon argues that platform power can be missed when competition is measured mainly through short-term price and output effects [10]. The Stigler Committee similarly emphasizes that digital platforms may generate durable market power through network effects, data advantages, and control over user access points [18]. In the European context, Cr  mer, de Montjoye, and Schweitzer argue that competition policy must account for the role of data, scale, and platform ecosystems in digital markets [3].

Recent AI-specific competition work extends these concerns to foundation models and AI infrastructure. The OECD identifies training data, deployment channels, compute, and cloud infrastructure as important stages in the AI value chain where concentration may affect competition [14, 15]. These works motivate the view that AI market power is not reducible to product-market share alone. Our work builds on this literature but shifts the object of analysis from market power over rivals to behavioral power over users.

2.2 Privacy, Surveillance, and Informational Power

Privacy scholarship has long argued that personal data collection creates asymmetries between individuals and institutions. Solove’s taxonomy of privacy harms distinguishes information collection, processing, dissemination, and invasion as analytically distinct privacy problems [17]. Nissenbaum’s theory of contextual integrity argues that privacy is violated when information flows breach context-relative norms rather than merely when information is disclosed [13]. Cohen further argues that networked information environments shape the conditions under which subjects form preferences, identities, and practical agency [2].

Zuboff’s account of surveillance capitalism describes behavioral data as a resource for prediction and behavioral modification [21, 22]. Pasquale’s work on black-box systems emphasizes how opaque algorithmic infrastructures can govern access to information, reputation, finance, and opportunity [16]. These works provide the normative and institutional background for our claim that the concentration of behavioral data threatens autonomy, not merely confidentiality.

2.3 Algorithmic Manipulation and Behavioral Influence

A related literature studies how digital systems can influence users' choices. Calo develops the concept of digital market manipulation, arguing that firms can use data-driven systems to exploit individual cognitive vulnerabilities [1]. Susser, Roessler, and Nissenbaum define online manipulation as covertly influencing decision-making by targeting vulnerabilities, and they identify autonomy as a central harm [19]. In empirical HCI research, Mathur et al. document dark patterns at scale across approximately 11,000 shopping websites and develop a taxonomy of interface-based manipulation [11].

This literature shows that manipulation is not merely hypothetical. However, most existing work analyzes particular manipulative interfaces, advertising systems, or recommender mechanisms. Our contribution is to connect manipulation capacity to data concentration itself. We argue that the more complete a firm's behavioral model of a user becomes, the more powerful and less visible its influence strategies may become.

2.4 AI Governance, Autonomy, and Emerging Regulation

Recent regulatory developments increasingly recognize manipulation and cross-context data combination as governance concerns. The EU Artificial Intelligence Act prohibits certain AI practices that deploy manipulative or exploitative techniques likely to cause significant harm [6]. The EU Digital Markets Act restricts designated gatekeepers from combining personal data across core platform services without valid consent [5]. In the United States, the Federal Trade Commission has warned that companies may act unfairly or deceptively if they retroactively expand data-use permissions, including for AI training, through quiet changes to terms of service or privacy policies [7].

These developments support the policy relevance of our framework. Yet existing law still tends to treat manipulation, privacy, and competition as separate regulatory domains. We instead model them as connected: data concentration increases predictive power, predictive power increases influence capacity, and influence capacity can reduce meaningful autonomy.

2.5 Privacy-Preserving Computation and Cryptographic Separation

Technical work on privacy-preserving computation provides tools for limiting data exposure while preserving computational utility. Differential privacy offers a formal framework for limiting what can be inferred about individuals from statistical outputs [4]. Federated learning trains models over decentralized data without requiring all raw data to be centralized [12]. Secure multiparty computation enables parties to compute functions over private inputs without revealing those inputs [20]. Fully homomorphic encryption allows computation over encrypted data [8], and zero-knowledge proofs allow verification of claims without revealing the underlying witness [9].

These tools are usually motivated by confidentiality, security, or privacy leakage. We reinterpret them institutionally: cryptographic separation can also function as an antitrust-adjacent governance

mechanism. Layer-wise encryption and compartmentalized access can prevent a single firm from reconstructing a complete behavioral model of a user, thereby bounding behavioral power.

2.6 Research Gap

Existing work provides strong accounts of digital market concentration, privacy harms, surveillance, algorithmic manipulation, and privacy-preserving computation. However, these literatures remain only partially integrated. Antitrust scholarship often focuses on competition among firms; privacy scholarship often focuses on information flows; manipulation scholarship often focuses on particular influence mechanisms; and cryptographic research often focuses on confidentiality.

This paper connects these domains through the concept of behavioral power. We argue that concentrated behavioral data can create a structural capacity to predict and influence users, and that preserving autonomy therefore requires both antitrust bounds on data concentration and technical constraints such as layer-wise encryption.

3 Behavioral Visibility and Behavioral Power

We distinguish two dimensions of platform power. The first is *behavioral visibility*: the degree to which a platform can infer a user's latent behavioral state from collected data. The second is *behavioral power*: the degree to which the platform can use this knowledge to change the user's future actions.

3.1 Latent User State

Let $x_u \in X$ denote the latent behavioral state of user u . This state is not directly observable. It represents a high-dimensional vector of behavioral features, including preferences, routines, beliefs, vulnerabilities, emotional responses, purchasing propensities, social ties, and attention patterns.

A platform observes only behavioral traces:

$$D_u = \{d_1, d_2, \dots, d_m\}.$$

Each observation may come from a different context, such as search, messaging, location, payments, health, education, or productivity software. The platform then constructs an internal estimate

$$\hat{x}_u = F(D_u),$$

where F may be a statistical model, recommender system, foundation model, or reinforcement-learning system.

The key governance concern is not simply that D_u contains private facts. Rather, the concern is that D_u allows the platform to infer unobserved properties of the user and to predict future conduct.

3.2 Behavioral Visibility

We define behavioral visibility as

$$M_u = \frac{I(D_u; x_u)}{H(x_u)}.$$

Here $I(D_u; x_u)$ is the mutual information between the platform's data and the user's latent behavioral state, while $H(x_u)$ is the entropy of the user's latent state.

This ratio normalizes informational access. If $M_u = 0$, the platform has no useful information about the user's latent behavioral state. If $M_u = 1$, the platform possesses complete behavioral visibility.

This definition has three important properties.

First, it is individual-specific. A platform may have high visibility over one user and low visibility over another.

Second, it is context-sensitive. The same amount of raw data may produce different visibility depending on whether the data spans many behavioral domains.

Third, it is model-dependent. Visibility increases not only when more data is collected, but also when better inference methods are developed.

Thus, behavioral visibility captures a form of informational power that traditional privacy metrics and market-share metrics do not measure.

3.3 Behavioral Power

Visibility alone does not imply control. A platform may predict behavior without changing it. We therefore define behavioral power separately.

Let π denote an intervention policy. This policy may include recommendations, rankings, advertisements, notifications, interface design, personalized prices, conversational responses, or content selection.

Let $x_u^{(0)}$ denote the user's behavioral trajectory without platform intervention, and let $x_u^{(\pi)}$ denote the trajectory under intervention policy π . Behavioral power is

$$S_u = \sup_{\pi} \mathbb{E} \left[d(x_u^{(0)}, x_u^{(\pi)}) \right],$$

where $d(\cdot, \cdot)$ is a distance function over behavioral trajectories.

This definition measures the maximum expected change in user behavior that the platform can induce. The supremum captures the fact that a platform may choose among many possible influence strategies.

The distance function d may be instantiated in different ways. For consumer behavior, it may measure changes in purchases. For political behavior, it may measure changes in beliefs or voting probability. For attention platforms, it may measure time allocation or engagement. For mental-health-sensitive contexts, it may measure changes in emotional or cognitive state.

Behavioral power is therefore not merely persuasion. It is the structural capacity to personalize interventions at scale using inferred behavioral knowledge.

4 A Cybernetic Model of AI Platforms

Modern AI platforms should be understood as feedback systems. They do not merely observe users once. They continuously observe, predict, intervene, and update.

Let $D(t)$ denote accumulated behavioral data at time t , $P(t)$ denote predictive capability, and $E(t)$ denote engagement or interaction intensity. We model the platform dynamics as

$$\frac{dD}{dt} = \alpha E,$$

$$\frac{dP}{dt} = \beta D,$$

$$\frac{dE}{dt} = \gamma P.$$

The first equation states that more engagement generates more data. The second states that more data improves predictive capability. The third states that better prediction increases engagement by improving personalization, relevance, or behavioral targeting.

Together, these equations form a positive feedback loop:

$$D \rightarrow P \rightarrow E \rightarrow D.$$

The parameters α , β , and γ have distinct interpretations. The parameter α measures the rate at which engagement is converted into data. The parameter β measures the rate at which data is converted into predictive capability. The parameter γ measures the rate at which predictive capability is converted into additional engagement.

4.1 Behavioral Reproduction Number

We define the behavioral reproduction number as

$$R_b = \alpha\beta\gamma.$$

This number summarizes the strength of the feedback loop. When

$$R_b > 1,$$

the platform is behaviorally self-reinforcing. In this regime, data collection produces prediction, prediction produces engagement, and engagement produces still more data.

This model explains why behavioral-data concentration can become self-amplifying. A firm that already possesses more behavioral data may improve prediction faster, attract more engagement, and collect even more data. This creates an endogenous tendency toward concentration even without traditional price-based exclusion.

5 Behavioral Concentration

Traditional antitrust analysis often uses market concentration measures such as the Herfindahl-Hirschman Index. However, revenue or output concentration does not capture concentration of behavioral knowledge.

Let M_i denote the behavioral visibility possessed by firm i . We define the Behavioral Herfindahl-Hirschman Index as

$$\text{BHBI} = \sum_i M_i^2.$$

A low BHBI indicates that behavioral visibility is distributed across multiple firms. A high BHBI indicates that one or a small number of firms possess most of the behavioral visibility over users.

The BHBI differs from conventional HHI in three ways.

First, the unit of analysis is not market share but informational share.

Second, the relevant resource is not sales, revenue, or output, but behavioral visibility.

Third, BHHI may be high even when product markets appear competitive, because different services may feed data into the same behavioral model.

For example, a firm may operate separate services in search, messaging, payments, cloud, and productivity. Each service may appear competitive in its own market, but the combined data may produce high behavioral visibility over the same user. BHHI is designed to capture this cross-context accumulation.

6 Autonomy Loss

We model autonomy as decreasing in behavioral power. Let A_u denote the autonomy of user u . We write

$$A_u = A_0 - \lambda S_u.$$

Here A_0 is baseline autonomy in the absence of platform influence, S_u is behavioral power, and λ measures the sensitivity of autonomy to behavioral intervention.

This formulation does not imply that all influence is illegitimate. Recommendations, reminders, and personalization may benefit users. Autonomy loss occurs when influence becomes sufficiently personalized, opaque, persistent, or asymmetric that users can no longer meaningfully distinguish their own preference formation from platform steering.

The parameter λ may vary across contexts. It may be higher for children, financially vulnerable users, emotionally distressed users, or users in high-stakes domains such as health, credit, employment, or politics.

7 Behavioral Domination Thresholds

We define behavioral domination as the joint presence of high visibility and high influence capacity:

$$M_u > M^*$$

and

$$S_u > S^*.$$

Both conditions are necessary. High visibility without intervention does not dominate users. High intervention capacity without visibility is less effective because it cannot be precisely personalized.

The threshold M^* represents the level of behavioral visibility above which the platform can infer sensitive or autonomy-relevant features of the user. The threshold S^* represents the level of influence capacity above which the platform can materially alter user behavior.

This threshold model is useful for regulation because it avoids treating all data collection or all personalization as harmful. Instead, it identifies the dangerous region where informational asymmetry and intervention capacity combine.

8 Cryptographic Compartmentalization

Cryptographic compartmentalization limits behavioral visibility by preventing any single organization from integrating all behavioral data layers.

Let user data be partitioned into layers:

$$D_u = \bigcup_{k=1}^n D_u^{(k)}.$$

Each layer corresponds to a behavioral domain, such as identity, communications, finance, location, health, productivity, or social relationships.

Each layer is encrypted separately:

$$E_k = \text{Enc}(D_u^{(k)}, K_k).$$

The key point is institutional separation. No single actor should possess all decryption keys $K_1, \dots, K_n$. Cross-layer access should require authorization, auditing, purpose limitation, and possibly multi-party approval.

Under unrestricted access, the platform observes

$$D_u^{\text{full}} = \bigcup_{k=1}^n D_u^{(k)}.$$

Under compartmentalized access, the platform observes only a subset:

$$D_u^{\text{sep}} = \bigcup_{k \in A} D_u^{(k)},$$

where $A \subset \{1, \dots, n\}$.

If the excluded layers contain information about x_u , then

$$I(D_u^{\text{sep}}; x_u) < I(D_u^{\text{full}}; x_u).$$

Therefore,

$$M_u^{\text{sep}} < M_u^{\text{full}}.$$

Cryptographic separation thus acts not merely as privacy protection, but as a structural limit on behavioral visibility.

9 Behavioral Antitrust

Behavioral antitrust aims to prevent excessive concentration of behavioral visibility and influence capacity.

Possible remedies include:

- restrictions on cross-service data combination,
- limits on behavioral-data mergers,
- data portability rights,
- interoperability mandates,
- structural separation between data collection and intervention systems,
- limits on exclusive access to behavioral datasets,
- independent audits of recommender and targeting systems,
- prohibitions on manipulative personalization in high-risk domains.

These interventions operate on different parts of the model. Limits on data combination reduce M_u . Interoperability and portability reduce lock-in and weaken the feedback parameter γ . Merger restrictions limit BHHI. Audits and prohibitions on manipulative design constrain S_u .

Thus, behavioral antitrust is broader than conventional antitrust. It does not only ask whether firms can raise prices or exclude competitors. It asks whether firms can accumulate sufficient behavioral knowledge and intervention capacity to threaten user autonomy.

10 Autonomy-Preserving Governance

The central governance result is that behavioral domination can be prevented by bounding either visibility or influence.

THEOREM 10.1 (AUTONOMY-PRESERVING GOVERNANCE). *Suppose cryptographic compartmentalization guarantees*

$$M_u \leq M_{\max},$$

and behavioral antitrust guarantees

$$S_u \leq S_{\max}.$$

If

$$M_{\max} < M^*$$

or

$$S_{\max} < S^*,$$

then behavioral domination cannot occur.

PROOF. Behavioral domination requires both

$$M_u > M^*$$

and

$$S_u > S^*.$$

If cryptographic compartmentalization ensures

$$M_u \leq M_{\max} < M^*,$$

then the visibility condition is false.

If behavioral antitrust ensures

$$S_u \leq S_{\max} < S^*,$$

then the influence condition is false.

Since domination requires both conditions simultaneously, preventing either condition is sufficient to prevent domination. $\square$

This theorem formalizes the complementarity between privacy-preserving technical architecture and antitrust-style structural governance. Cryptography limits what platforms can know. Antitrust limits what platforms can consolidate and do.

11 The Limits of Ex-Post Governance

Many governance mechanisms operate after data has already been collected and models have already been trained. These include transparency reports, explainability tools, audits, disclosure obligations, and notice-and-consent frameworks.

Such mechanisms may improve accountability, but they do not necessarily reduce behavioral visibility.

11.1 Information Monotonicity

Suppose a platform has collected data D_u and constructed

$$\hat{x}_u = F(D_u).$$

If a governance mechanism G does not alter the stored data, learned parameters, or future inferential capability of the platform, then the platform's informational position remains unchanged.

LEMMA 11.1 (INFORMATION MONOTONICITY). *Let G be a governance mechanism that does not remove data, erase model parameters, restrict inference, or prevent future recombination. Then*

$$I(D_u; x_u) = I(G(D_u); x_u).$$

Consequently,

$$M_u = M'_u.$$

PROOF. The mechanism G may disclose information to users or regulators, but it does not reduce the information available to the platform. Since the platform retains the same data and inferential capacity, its mutual information with the user's latent state remains unchanged. Therefore behavioral visibility is unchanged. $\square$

11.2 Impossibility of Purely Ex-Post Protection

THEOREM 11.2 (EX-POST GOVERNANCE IMPOSSIBILITY). *Consider a platform possessing behavioral visibility M_u . Suppose governance mechanisms satisfy the following conditions:*

- (1) *they do not reduce stored behavioral data;*
- (2) *they do not reduce learned model parameters;*
- (3) *they do not restrict future inference;*
- (4) *they do not prevent cross-context recombination.*

Then such mechanisms cannot guarantee

$$M_u < M^*.$$

Consequently, purely ex-post governance cannot guarantee autonomy preservation.

PROOF. By Information Monotonicity, mechanisms that leave data, model parameters, and inference capacity intact do not reduce behavioral visibility. Therefore, if

$$M_u > M^*$$

before the governance intervention, then

$$M_u > M^*$$

after the intervention. Since behavioral domination requires excessive visibility together with influence capacity, a mechanism that cannot reduce visibility cannot universally prevent domination. $\square$

12 Structural Governance

Structural governance acts before excessive behavioral visibility is formed. It changes the architecture of data collection, storage, combination, and use.

Examples include:

- purpose limitation at the data-layer level,
- cryptographic separation of behavioral domains,
- legal restrictions on cross-context data fusion,
- mandatory data minimization,
- separation between data controllers and intervention systems,
- limits on behavioral-data acquisitions,
- interoperability requirements that reduce lock-in,
- deletion and machine-unlearning duties.

Unlike ex-post mechanisms, structural mechanisms directly alter the platform’s informational state. They reduce D_u , restrict F , or limit the intervention policy space Π .

In the notation of this paper, structural governance is effective when it ensures either

$$M_u \leq M_{\max} < M^*$$

or

$$S_u \leq S_{\max} < S^*.$$

13 Principle of Autonomy Preservation

The framework yields the following principle:

No institution should simultaneously possess sufficient behavioral visibility and sufficient intervention capability to achieve behavioral domination over individuals.

This principle unifies privacy, antitrust, cryptography, and AI governance. Privacy limits unnecessary observation. Cryptography limits data recombination. Antitrust limits structural concentration. Interoperability reduces lock-in. Auditing constrains hidden steering.

The central policy target is therefore not only market power or privacy loss, but the joint concentration of knowledge and influence.

14 Conclusion

The central governance challenge of AI is not merely privacy, but concentration of behavioral power.

As AI systems gain access to increasingly comprehensive behavioral datasets, they may acquire unprecedented capacities to predict and shape human action.

We propose a theoretical framework in which autonomy loss emerges from excessive data concentration. The framework motivates both antitrust limits and cryptographic compartmentalization.

Future work should develop empirical estimates of behavioral power, identify concentration thresholds in real-world platforms, and design practical regulatory mechanisms for autonomy-preserving AI governance.

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